

Townsend 1.1-1.3

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Questions for Gallicchio

Why does the phase change from low to high? I've seen the demonstration on strings and believe it, but what is the intuitive story explanation?

What determines the behavior for wave-particle duality? What systems do we observe one/the other/ both?

I don't really understand Compton scattering - so you both have an electromagnetic wave moving electrons which then makes another wave(the expected result) and also a wave generated by bouncing photons off of electrons (the angle depends on the frequency? the impact parameter?), and that bounce transfers momentum and shifts the energy of the photon, so bigger scatter = more energy loss = lower frequency = higher wavelength?

Waves

$$\begin{aligned}k &= \frac{2\pi}{\lambda} \\ \omega &= \frac{2\pi}{T} = 2\pi\nu \\ \frac{k\lambda}{4} &= \frac{\pi}{2}\end{aligned}$$

So adding a translation of $\frac{\lambda}{4}$ to x gives

$$\sin\left(k\left(x + \frac{\lambda}{4}\right) - \omega t\right) = \cos(kx - \omega t)$$

Maxwell's equation for electric field in free space:

$$\frac{\partial^2 \mathcal{E}}{\partial x^2} - \frac{1}{c^2} \frac{\partial^2 \mathcal{E}}{\partial t^2} = 0$$

This looks like the diff EQ that we got when we rearranged the Schrödinger eq, which makes sense because both describe wave motion.

The acceleration of the wave in space is related to the acceleration in time. Units cancel with c .

Cos and Sine are solutions to maxwell's eq. if $\omega = kc$.

$$k = \frac{2\pi}{\lambda}, \omega = \frac{2\pi}{T}$$

so

$$\frac{2\pi}{T} = \frac{2\pi}{\lambda} c \implies \frac{\lambda}{T} = c, \lambda\nu = c$$

In a medium with index of refraction n , the speed of light becomes $\frac{c}{n}$. The frequency of the light stays the same, so the wavelength must decrease to compensate. It is traveling slower but oscillating the same speed, so the peaks would be closer to each other in space.

1.2 Interference and Diffraction

Light going from low index of refraction has a phase change when reflected, but the wave continuing is the same. Going from high to low, it continues how it is because some energy is absorbed but it isn't slowed down.

Why does the phase change from low to high? I've seen the demonstration on strings and believe it, but what is the intuitive story explanation?

Light going through a slit is usually paired between each half of the slit. If the path length difference $d = a \sin \theta = \lambda$, they interfere.

I was imagining splitting the slit into a prime number of divisions, but this wouldn't cause destructive interference - just stacking waves that are at different phase shifts. The interference comes when things are paired. I answered my own question.

Photons

If we take a photograph with vanishingly small amounts of light to activate the reaction, instead of decreasing the image uniformly we end up with stark spots of light. This is counter to the wave behavior, and is more like small particles hitting at one spot or another.

What determines the behavior for wave-particle duality? What systems do we observe one/the other/ both?

Compton scattering

If you shine light on a block of carbon, you would expect the electrons to have a system response to the light, and have the same steady state frequency. However, we also see a secondary dominant frequency that depends on the scattering angle. The photon bounces off the electron (and that bounces the other way).

The shift in wavelength is $\frac{h}{mc}$, where h is plank's constant, and m is the mass of the object (i.e. electron).

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